How LLMs generate tokens, and how to prompt them

Lesson 1 · LLM Fundamentals, Prompting & API Integration

By the end of today, you can

- **Explain**, on a diagram, how a transformer turns text into next-token prediction
- **Count** a prompt's tokens and **price** a call
- **Explain** why output drives cost, and why a token cannot be taken back
- **Tell** a model's own thinking from reasoning you ask for
- **Choose** a prompt pattern, and **measure** it on a test set
- **Write** a system prompt that fixes the answer's format

Albert's Marketplace: the support desk

- Messages in **six countries**: parcels, refunds, double charges, sellers, products
- **Routed by hand** today; a misrouted message waits days
- **You are the LLM team:** route each message, automatically, and prove it is good enough

Tickets	Labels
120, each on a real order	billing · delivery · returns_refunds · account · seller · product · information · other

The same ticket, 40% dearer

Ticket T020	Words	Tokens (Claude Haiku 4.5)
English	103	161
French	110	**234 (+45%)**
German	103	**229 (+42%)**

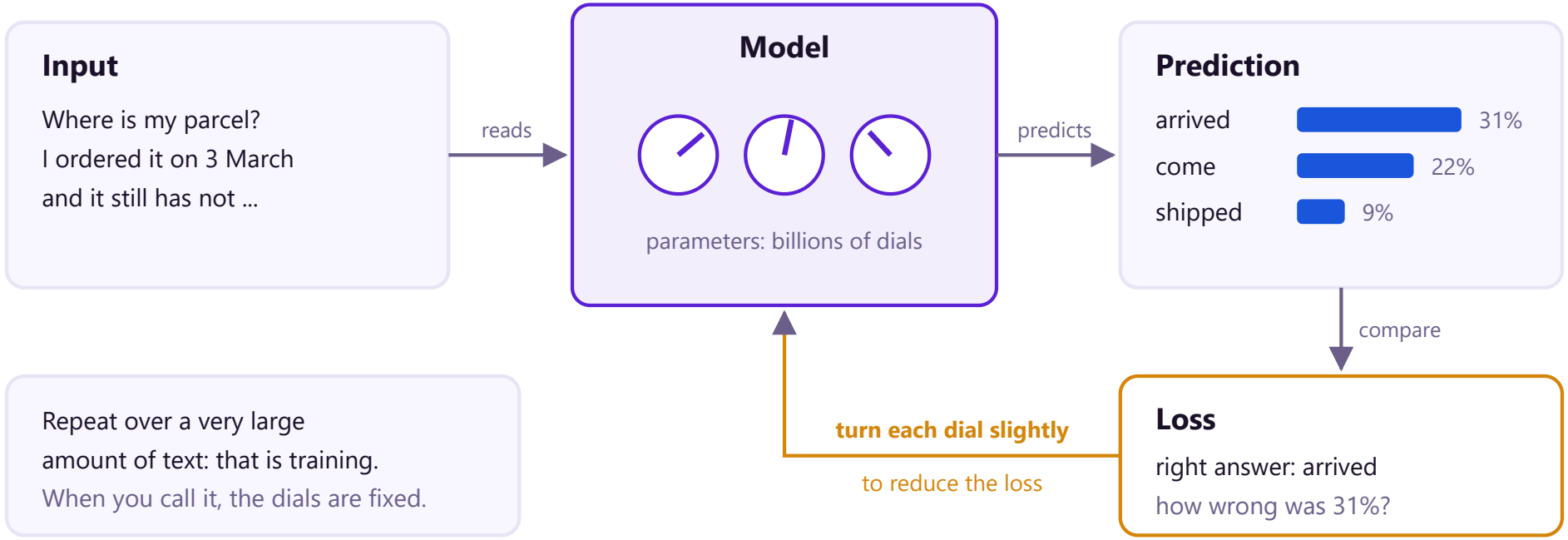
Measured 1 October 2026. **Why?** A model does not read words.

1 · What a language model is

A model is a function with parameters

- **Parameters:** billions of numbers that set its behaviour
- **Training:** predict, measure the **loss**, nudge every parameter
- **At use time:** the parameters are fixed. It computes; it does not learn

A model is a function; training turns its dials



Percentages are illustrative: the shape of the idea, not one model's output.

The training task: the next token

> Where is my parcel? I ordered it on 3 March and it still has not ...

- The data holds the true continuation: *arrived*
- One task, at enormous scale → translation, summary, classification

Tokenization: text becomes token ids

Characters	Tokens	Vocabulary
104	25	200,019

tiktoken, o200k_base vocabulary
(OpenAI's open tokenizer)

The text, one colour per token

Hi | , | ·I | ·ordered | ·the | ·Sally | ·Hansen | ·Miracle | ·Gel | ·Set | , | ·order | ·number | · | 181 | 571 | . | ·Could | ·you | ·check | ·where | ·my | ·parcel | ·is | ?

What the model receives: the token ids

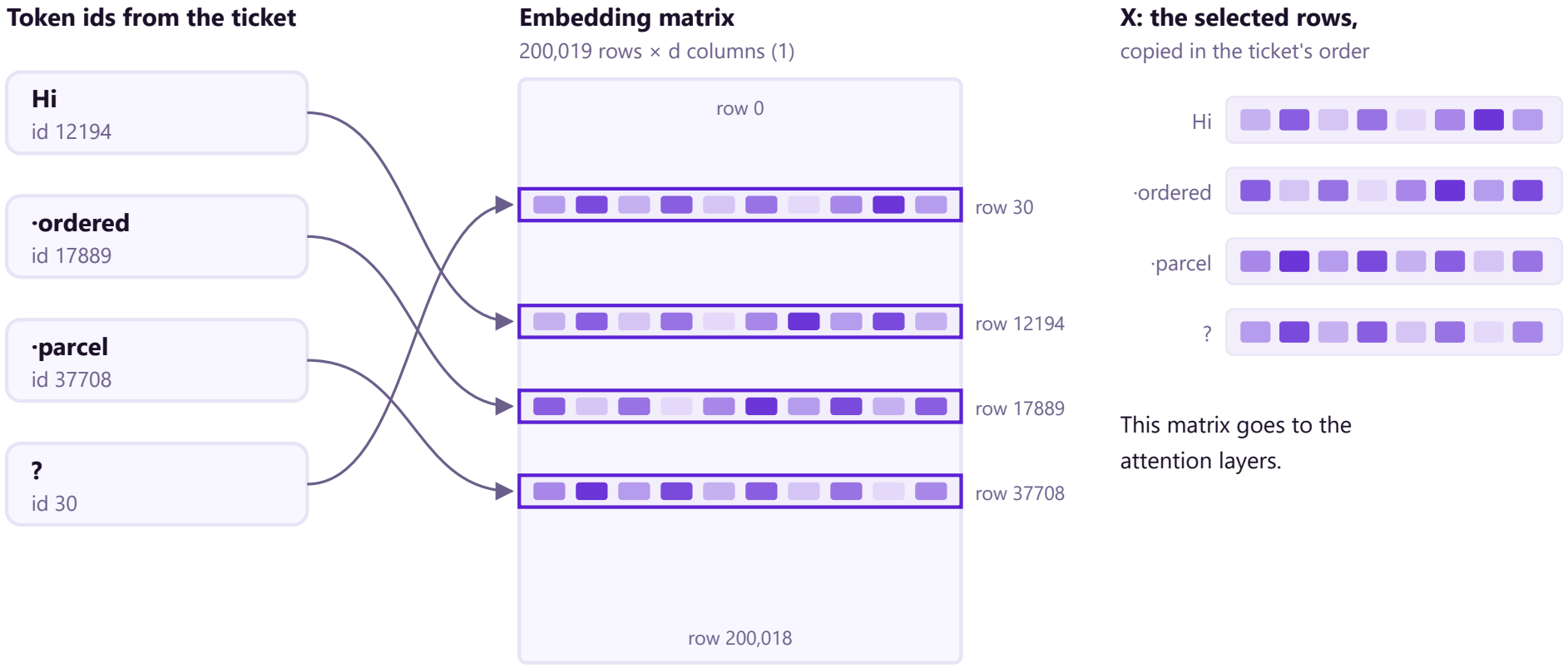
12194 | 11 | 357 | 17889 | 290 | 88213 | 88786 | 120276 | 28986 | 3957 | 11 | 2569 | 2086 | 220 | 16813 | 41467 | 13 | 30456 | 481 | 2371 | 1919 | 922 | 37708 | 382 | 30

· marks a space: most tokens carry the space before the word. The order number 181571 becomes " ", "181" and "571". Claude's own tokenizer is not published; its cut differs in detail, not in kind.

Embeddings (optional)

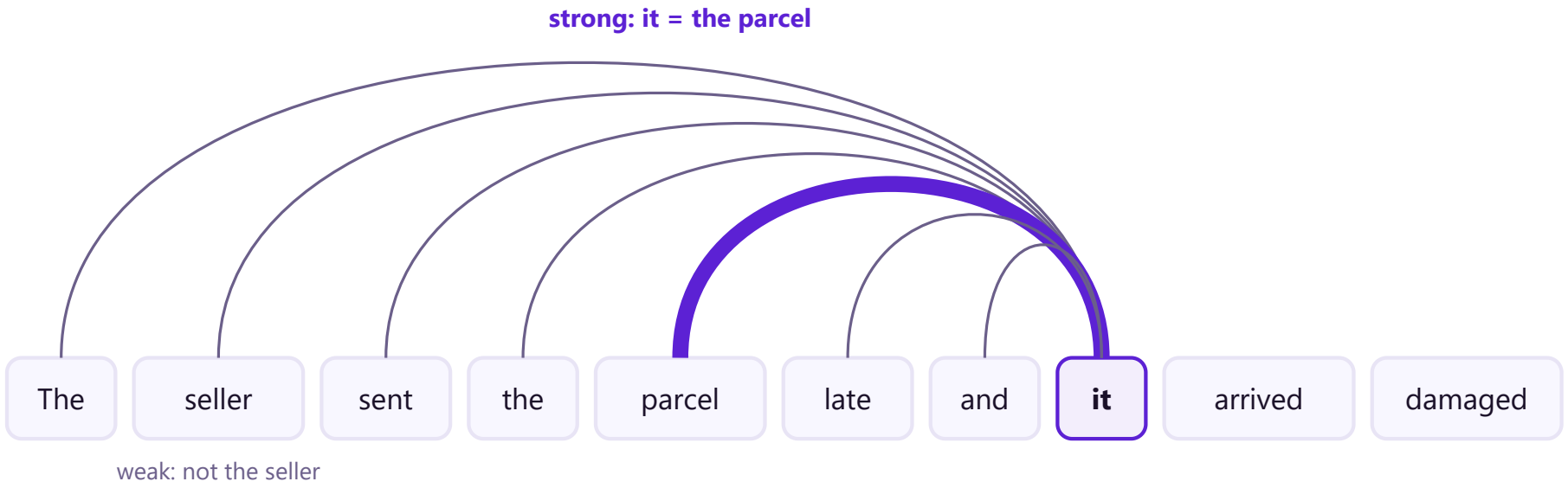
- **Embedding matrix:** one row per vocabulary entry, learned
- A token id is a **row number**; its embedding is that row
- Original transformer: rows of **512** numbers
- Similar uses → similar vectors (lesson 3)

Embeddings: a lookup in a learned table



(1) d is the length of each vector: 512 in the original transformer (Vaswani et al., 2017). Claude's is not published. The rows' numbers are learned in training; the shades shown are illustrative. Token ids: tiktoken, o200k_base.

Attention: which earlier words does "it" use?



At every layer, each token gathers information from the tokens before it, weighted by how much each one matters to it.

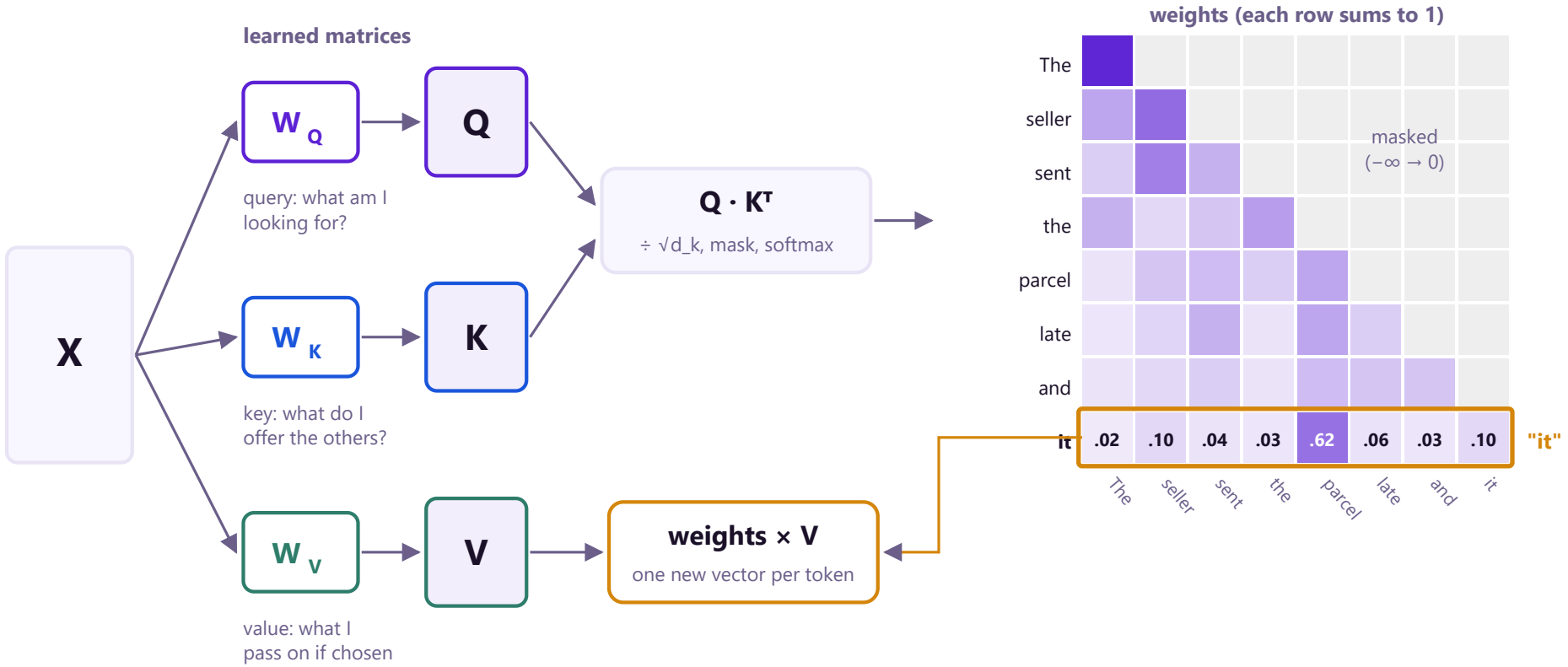
Line thickness is illustrative: models do not publish these weights for a given sentence.

Queries, keys and values (optional)

Vector	Answers
query	what am I looking for?
key	what do I offer?
value	what do I pass on?

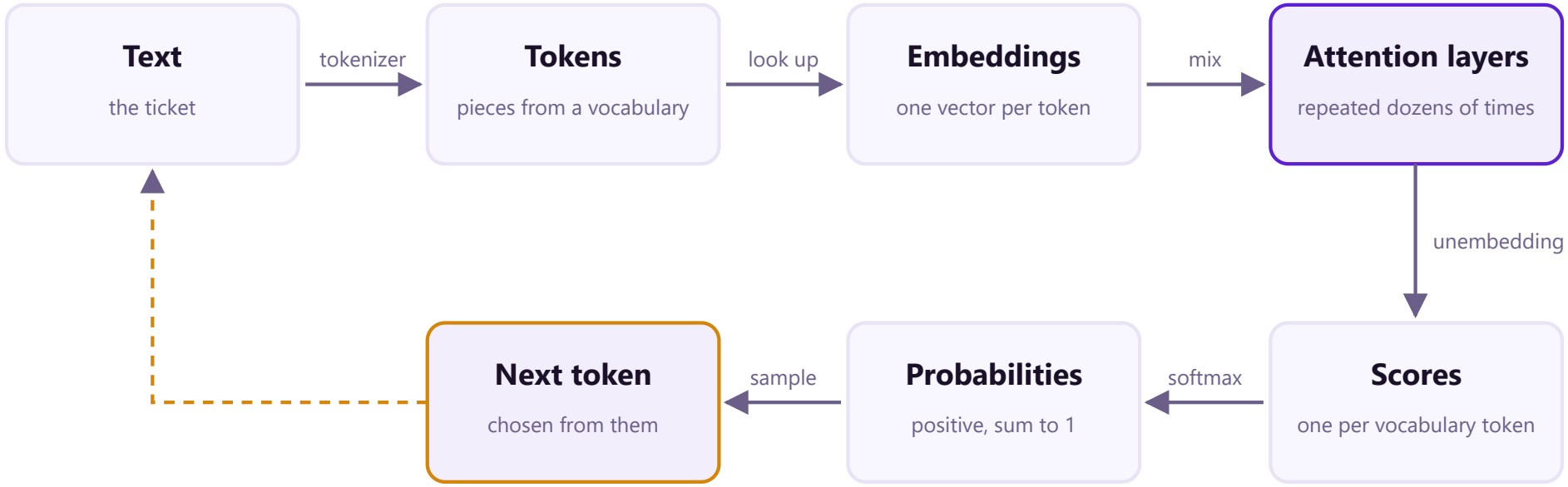
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

Attention: queries, keys and values



d_k: the length of a key vector (64 per head in the original transformer). T: transpose. Masking sets the scores of later tokens to −∞, so their weights are 0. Weights are illustrative; the "it" row matches the attention figure.

From a ticket to the next token

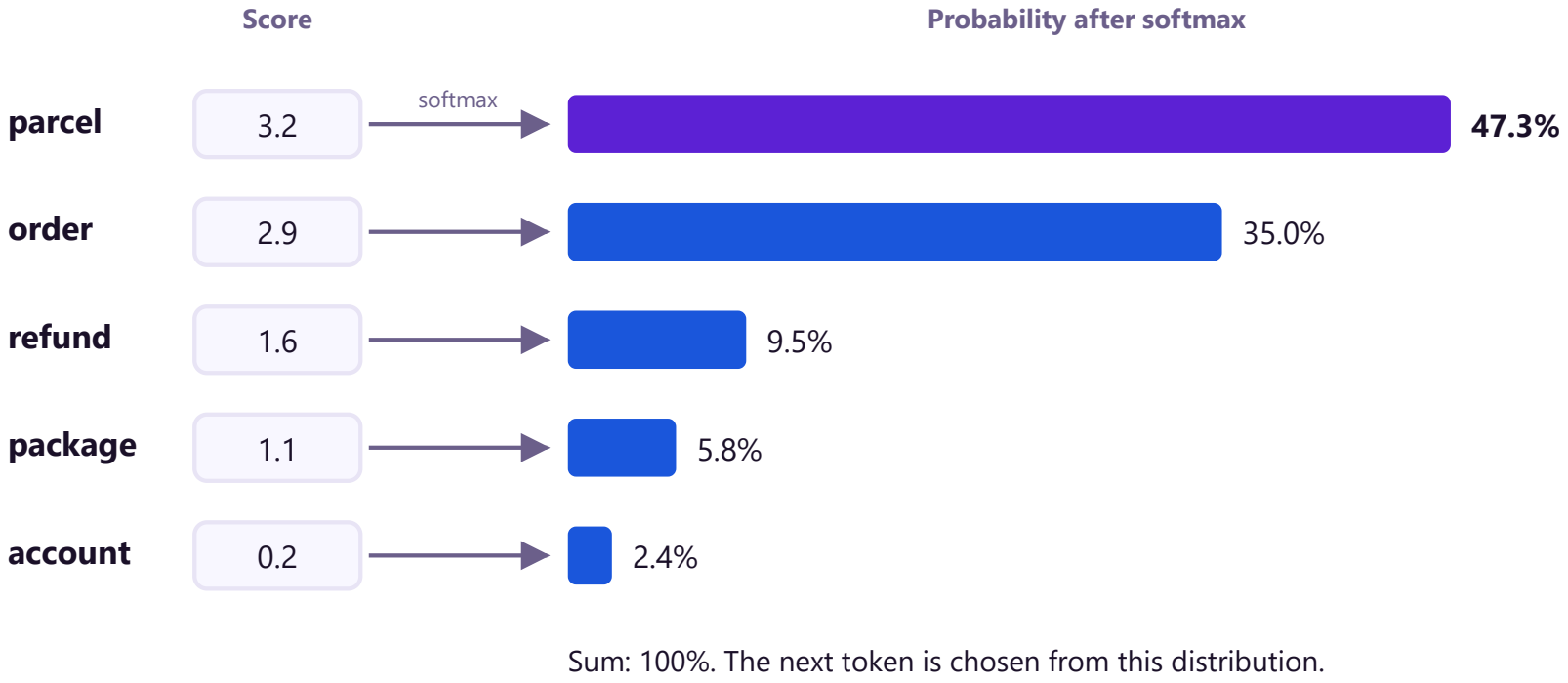


append the chosen token to the input, and run again

One pass of this loop writes one token of the answer.

Softmax: scores become probabilities

After "Where is my", five candidate next tokens:



Sum: 100%. The next token is chosen from this distribution.

Scores are illustrative: Claude does not return them. The probabilities are computed from these scores.

From a base model to an assistant (optional)

- **Base model:** continues text. Ask a question, get more questions
- **Instruction tuning:** examples of instructions followed well
- **RLHF** (reinforcement learning from human feedback): people rank answers
- → why *"Answer with the label only"* works at all

2 · Tokens, and what they cost

A tokenizer splits text into a fixed vocabulary

Word, on its own	Tokens (tiktoken)
refund	`refund`
remboursement	`rem` · `b` · `ourse` · `ment`
Rückerstattung	`R` · `ück` · `erst` · `attung`

Frequent in the training text → one token. Mostly English.

The same message, three languages, three prices

One message, cut by an open tokenizer (tiktoken, o200k_base). Each chip is one token.

		Claude Haiku 4.5 (1)
English	Where ·is ·my ·parcel ? ·I ·would ·like ·a ·refund . 11 tokens here	**20 tokens** $0.20 per 10,000
French	O ù ·est ·mon ·colis ·? ·Je ·voudrais ·un ·remboursement . 11 tokens here	**24 tokens** $0.24 per 10,000
German	Wo ·ist ·mein ·Paket ? ·Ich ·möchte ·eine ·Rück erst attung . 12 tokens here	**29 tokens** $0.29 per 10,000

· marks a space: most tokens carry the space before the word.
(1) Counted by Anthropic's token-counting endpoint, 1 Oct 2026, including the message's own framing; priced at $1 per million input tokens. Claude's tokenizer is not published, hence the open one above.

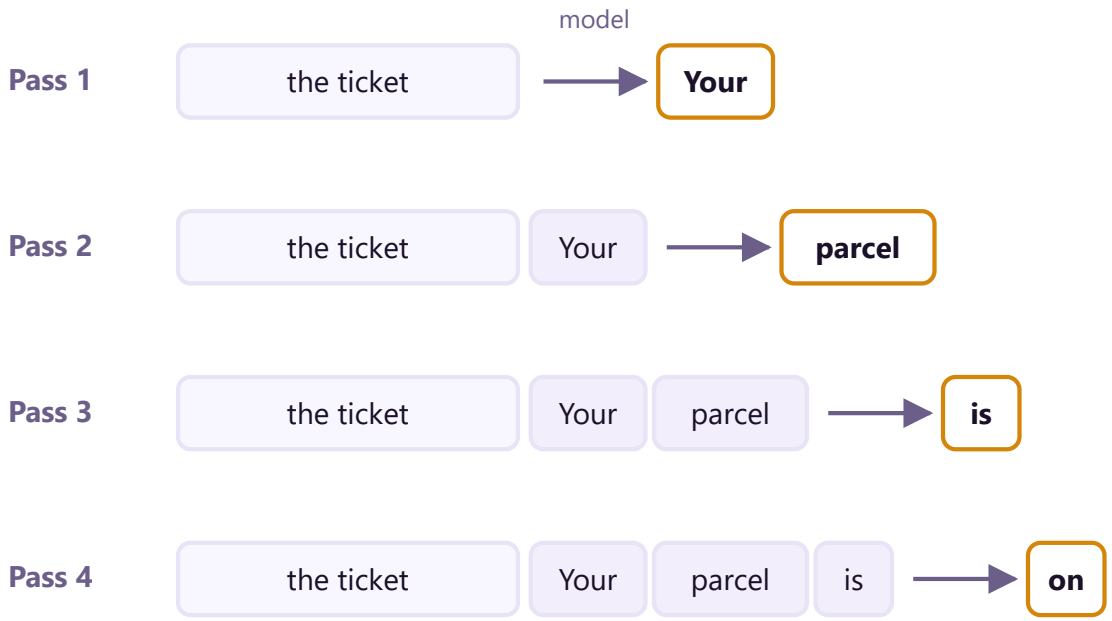
Count with the model you are billed on

Same English ticket	Tokens
tiktoken	134
Claude Haiku 4.5	161
Claude Opus 5.5	**217**

Anthropic's token counting: **free**

3 · Decoding: one token at a time

Decoding: one token per pass



Reading the prompt: one pass over all its tokens at once.

Writing the answer: one full pass per token.

Output costs 5× input

Haiku 4.5, Sonnet 5.5, Opus 5.5 (1)

(1) Anthropic pricing page, read 1 Oct 2026. The answer shown is illustrative.

One-word answers, two bills

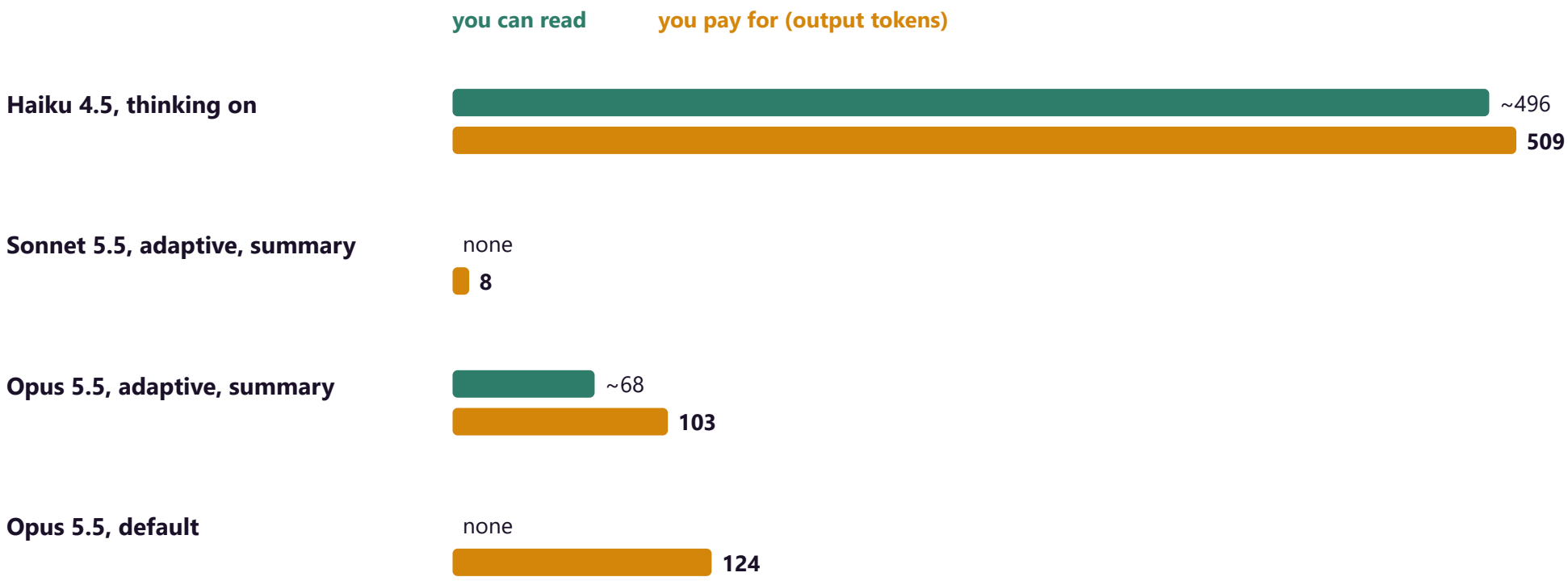
"Answer with the label only", ten tickets:

	Haiku 4.5	Opus 5.5
Output tokens	**58**	**487**
Cost	$0.0039	$0.0287

A token, once written, cannot be taken back → reason first, answer last

4 · Model thinking

Thinking: what you read, what you pay



Same ticket (T001), "answer with the label only". All four answered returns_refunds.

Measured 1 Oct 2026. "You can read": tokens in the returned thinking text, counted with the same model.

In the prompt, or in the model

	In your prompt	The model's own thinking
Where	the answer you read	a separate `thinking` block
Readable	always	some models; a summary or nothing on others
Billed	as output	as output, **even unseen**

5 · The context window

The context window, and where the answer sits

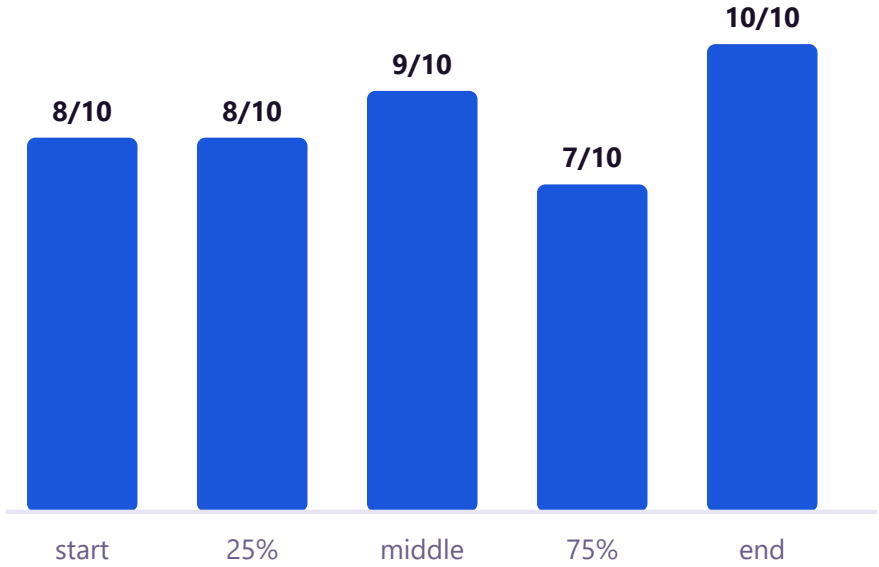
Everything in one request counts



Context window, from the Models API (1):

Haiku 4.5	**200,000 tokens**
Sonnet 5.5	**1,000,000 tokens**
Opus 5.5	**1,000,000 tokens**

Is the middle read worse? (2)



No dip in the middle. Every error picked a look-alike ticket, not a far-away one.

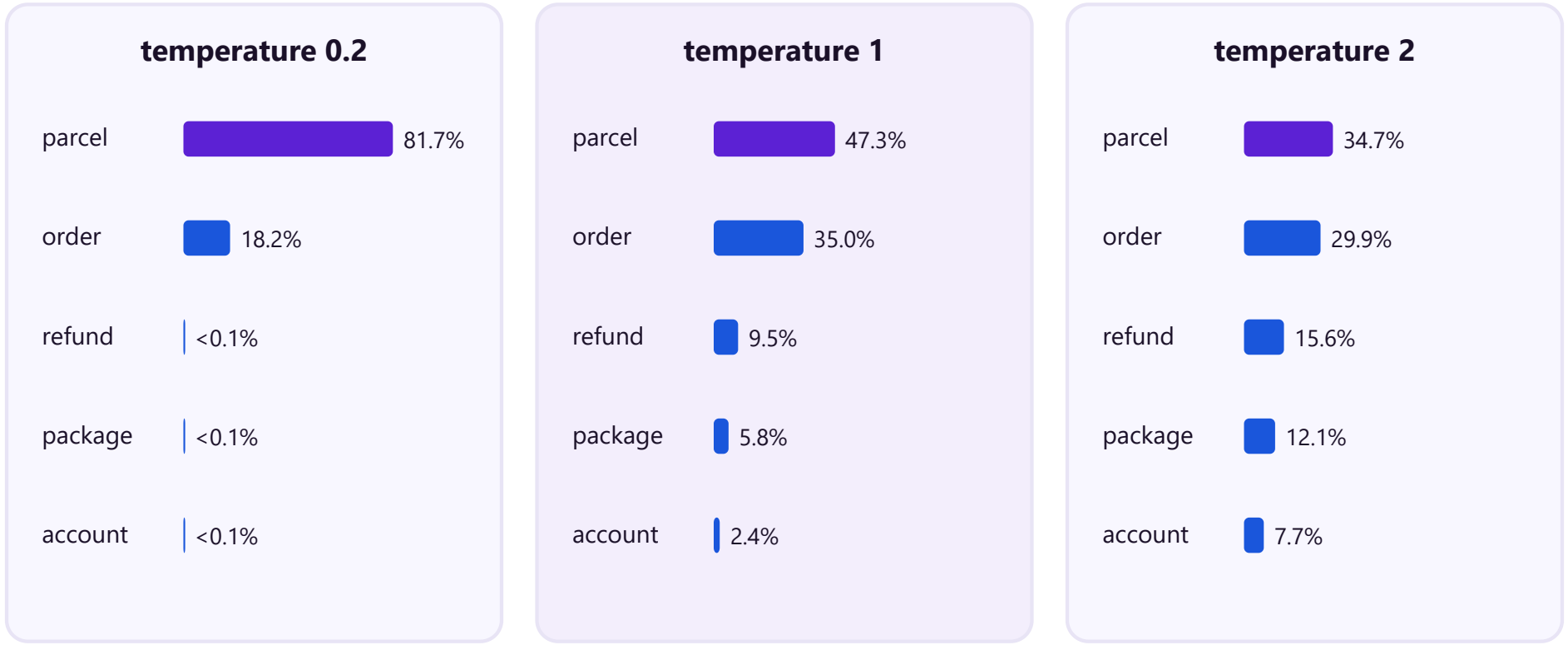
(1) Anthropic Models API, 1 Oct 2026. Maximum output: 64,000 (Haiku), 128,000 (Sonnet, Opus).
(2) Haiku 4.5, one prompt of 28,795 tokens (120 tickets), the target ticket placed at five positions, ten targets, 1 Oct 2026: about 14% of Haiku's window, so not a long-context test.
Liu et al. (2023) found a dip in the middle on 2023 models.

Lost in the middle?

- **Liu et al. (2023):** a dip in the middle, on 2023 models
- **Ours:** 8 · 8 · 9 · 7 · 10 out of 10. **No dip**
- Every error: **a look-alike ticket**
- 28,795 tokens = **14% of Haiku's window**: not a long-context test

6 · Sampling

Temperature reshapes the distribution



Below 1: sharper, the likely token wins more often. Above 1: flatter, more variety.

Computed from the illustrative scores of the softmax figure. Not accepted at 0 by Sonnet 5.5 or Opus 5.5 (1 Oct 2026).

What the models still accept

	`temperature: 0`	`temperature: 1`	`top_p: 0.9`
Haiku 4.5	yes	yes	yes
Sonnet 5.5	no	yes	no
Opus 5.5	no	yes	no

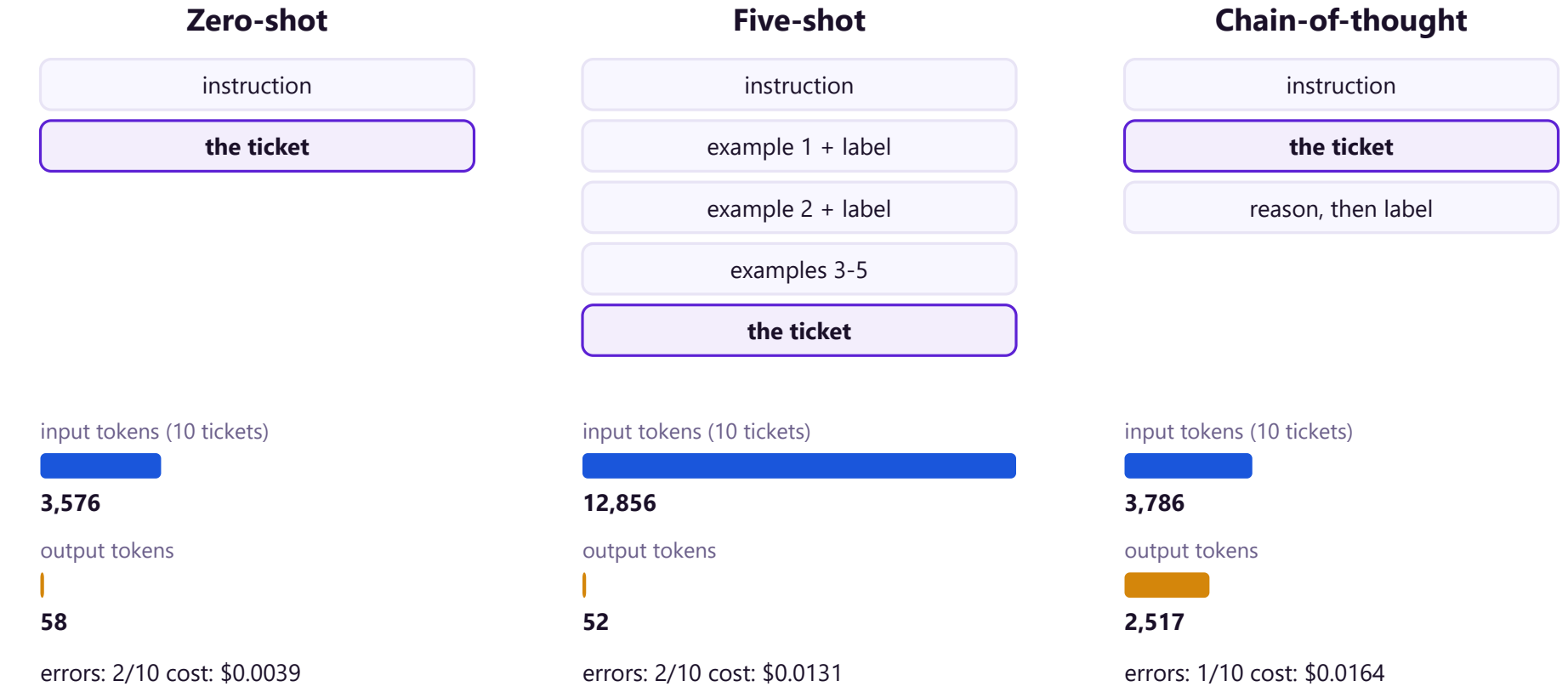
Consistency comes from the prompt, and is measured, not dialled.

7 · Three prompt patterns

Choose by what the task needs

Pattern	What it adds	When
Zero-shot	nothing	clear classification, recall
Few-shot	solved examples	overlapping categories
Chain-of-thought	reasoning first	several steps

Three ways to ask, three bills



Claude Haiku 4.5, ten tickets, 1 Oct 2026. Errors count the ticket's label or its second acceptable label.
Bars share one scale. On ten tickets, one error is ten points: none of these differences is significant.

8 · The system prompt

One sentence about the answer's format

With a format rule

system

Labels: billing, delivery, ... other.
... Answer with the label only.

user

five examples, then the ticket

↓

assistant

product

parser reads: product

Opus 5.5: 1 error out of 10

Without one

system

Labels: billing, delivery, ... other.
(no rule about the answer)

user

five examples, then the ticket

↓

assistant

Label: product
The core issue is ... the primer arrived expired. ... This isn't a seller case ...

the model said product; the parser took the last label word: seller

Opus 5.5: 7 errors out of 10

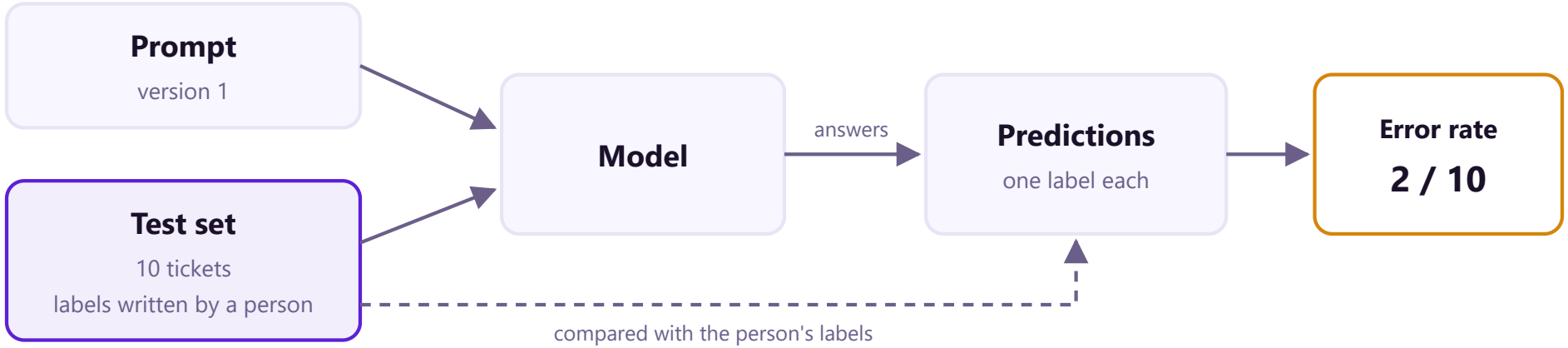
Claude Opus 5.5, five-shot, ten tickets, 1 Oct 2026; only the format rule changed. A second run on another ten tickets gave 2 against 3: the size varies, the mechanism does not. Right: a real answer (ticket T042), shortened.

The model was right. The measurement was wrong.

- No format rule → an explanation, several label words
- The parser took the **last** one
- 1 → 7 errors on one run; 2 → 3 on another. **Same mechanism**
- Rules in `system`, the ticket in `messages`: also the first defence against injection (lesson 3)

9 · Measuring a prompt

Measuring a prompt



What counts as wrong is a decision:

against the main label only: 4 / 10

or its second acceptable label: 2 / 10

Say which one you used.

Every figure carries four facts

model	claude-haiku-4-5
date	2026-10-01
parameters	zero-shot, default temperature
test file	10 tickets, tickets-v1.0.0

Three things to say, every time

- **Which rule:** strict, or a second acceptable label (4 vs 2 errors)
- **How many:** ten tickets → one error is ten points
- **The four facts:** model · date · parameters · test file

Now

1. **The demo:** the SDK, and every number of today, live
2. **The quiz:** twenty questions, twenty minutes
3. **Exercise 1:** your three prompts, on the test set, in pairs
4. **At home:** Exercise 2, where do the eight labels fail? And the project's first step